Aim

The goal was to build a circuit that can detect how many times 1011 appears in an incoming bit stream, even when sequences overlap with each other. The count is shown using a 7 segment display connected to a 4 bit counter.

Approach

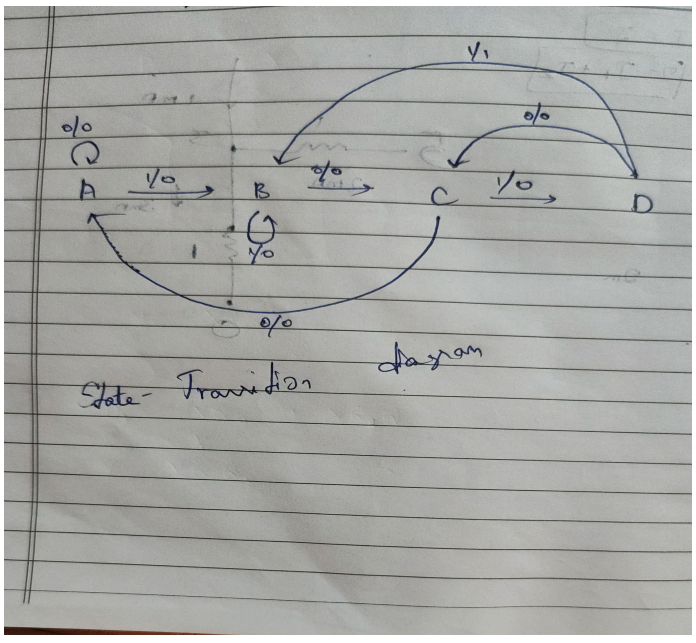
I honestly had no idea where to start since I hadn't done anything like this before. The first thing I did was watch some videos on FSMs and how Proteus works. Then I learned about Karnaugh maps which helped me simplify the boolean expressions.

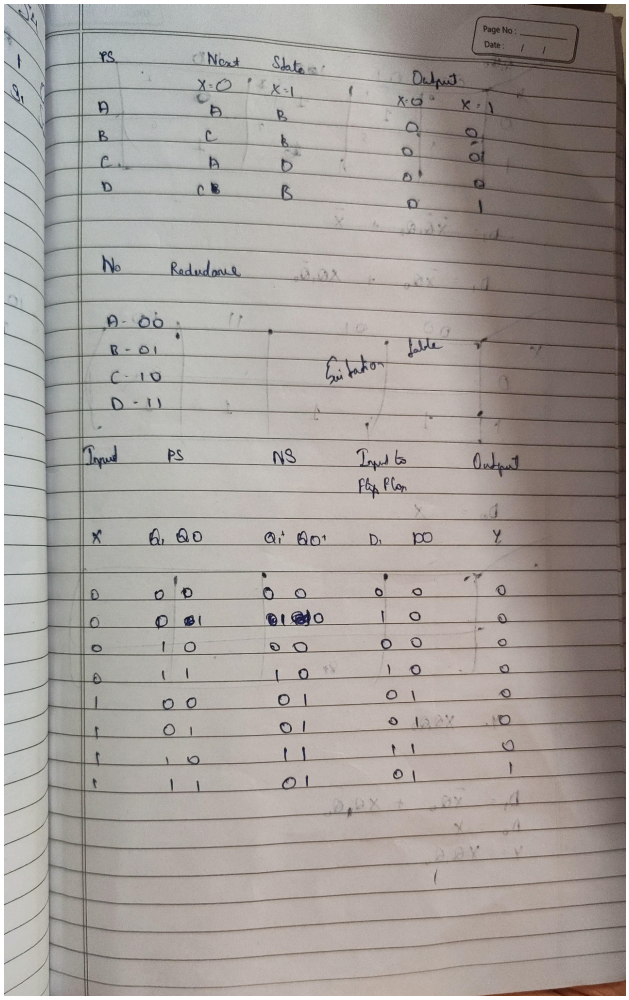
After that I drew the state transition diagram on paper with 4 states A, B, C, D — each representing how much of 1011 has been matched. From that I made the excitation table and got these expressions:

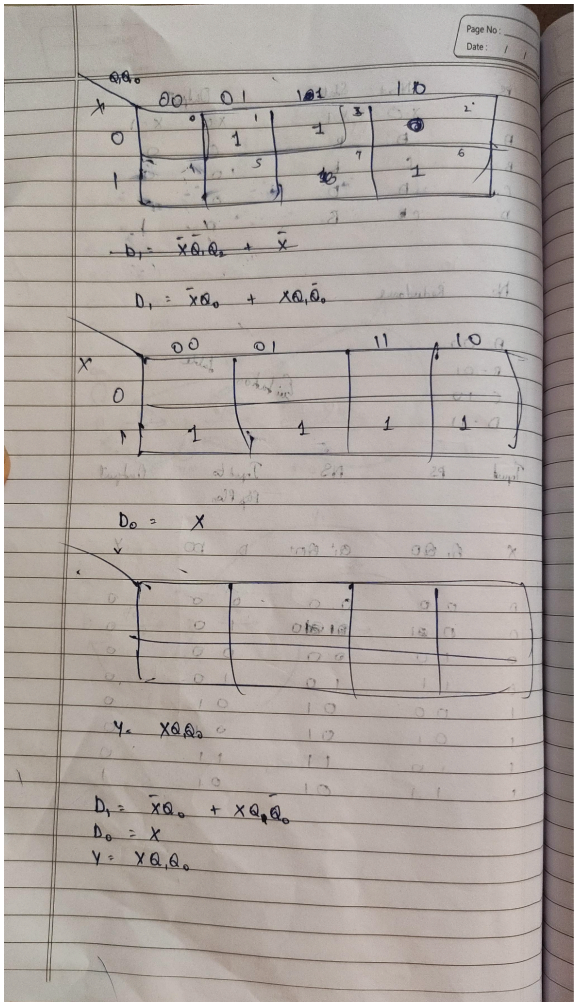
- $D1 = \bar{X}Q0 + XQ1\bar{Q}0$
- $D0 = X$
- $Y = X \cdot Q1 \cdot Q0$

I'll attach the handwritten working below.

Then I started building in Proteus. The gate and flip flop part came together relatively quickly but the counter part was a nightmare.







What Failed

The FSM itself was detecting correctly — I verified this by putting a logic probe on Y and it blinked exactly 3 times for the given sequence. But the 7 segment display was showing completely wrong values like C, 8, 4 instead of 3.

I thought the counter chip (74HC163) was broken or wrongly wired so I spent almost a full day trying different things — swapping to a 7490, adding flip flops in between, trying different clock configurations. None of it worked.

How I Fixed It

After placing logic probes on every single output pin of the counter I realized the counter binary outputs were actually correct all along. The problem was that I had connected the 7 segment

display inputs in the wrong order — Q3 was going to A and Q0 was going to D instead of the other way around. Once I swapped them the display immediately showed 1, 2, and 3.

Extra Features

- Used DPATTERN to feed the input automatically instead of clicking manually
- Verified with the given test input (1 0 1 1 0 1 1 0 0 1 0 1 1 0) and got 3
- Also tested with a few other sequences and verified manually that the count was correct each time

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Scope of Development

The circuit works but it looks messy. If I had more time I'd reorganize the layout, add proper labels to each block so it's easier for anyone reading it to understand what each part does without me having to explain it.
